

1.0 INTRODUCTION

1.1 Overview

1.1.1 The loading methods in individual closed car loading publications issued by AAR's Damage Prevention and Loading Services are minimum standards that have been evaluated and approved by the AAR Damage Prevention and Freight Claim Committee. The minimum standards offer practical guidelines on the subjects covered. Because these are minimum standards, it may be necessary to supplement the methods in some instances. In addition, each carrier may have its own requirements. Contact your origin carrier for more information.

1.1.2 Compliance with the "Minimum Loading Standards" contained herein will ensure conformance with Circular No. 42-N rules and provide adequate protection for lading from sources of damage in the normal railroad environment.

1.1.3 The general rules contained in Circular No. 42-N or supplements thereto issued by the AAR are formulated for the purpose of providing safe methods of loading boxcars and must be observed.

1.1.4 The loading rules and/or practices apply to shipments transported in the USA, Canada, and Mexico.

1.1.5 Securement standards in AAR closed car loading publications are intended for safe transit of the railcar from origin to destination and for the prevention of lading and equipment damage. The standards do not address unloading practices.

1.1.6 Loading and bracing methods not currently approved may receive consideration for approval and publication under the Damage Prevention and Loading Services General Information Bulletin, No. 2, "Procedures Governing Evaluation and Acceptance of New Closed Car Loading and Bracing Methods and Materials." Submit requests to Committee Manager, Damage Prevention and Freight Claim Committee, Association of American Railroads, Transportation Technology Center Inc., 55500 DOT Road, Pueblo, CO 81001. Alternatively, contact your serving carrier.

1.1.7 CAUTION: Car rocking motion caused by lift equipment entering and/or exiting the railcar may cause unsupported packages or articles with a high center of gravity to fall to the floor. Minimize access to the car. Exercise caution when inside a partially loaded car. Lift operators should stay on lift equipment, whenever possible, while inside a partially loaded car.

1.2 Reference Documents

NOTE: For the latest updates to these reference documents go to www.aar.com/standards/damage-publications.php.

1.2.1 Circular No. 42-N (or supplements thereto)

"General Rules Governing the Loading of Carload Shipments of Commodities in Closed Cars"—These requirements must be observed in all closed car loading activities to ensure safe transit of the railcar from origin to destination, thereby eliminating hazard to railroad operation.

1.2.2 Circular No. 43-G (or supplements thereto)

"Rules Governing the Loading, Blocking, and Bracing of Freight in Closed Trailers and Containers for TOFC/COFC Service"—This publication contains the requirements covering loads in trailers or containers.

1.2.3 *Closed Car Loading Guide*, Part 1, "Minimum Loading Standards for Freight in General Purpose Boxcars"—This publication provides minimum loading standards for freight in closed railcars.

1.3 Rail Transportation Environment

1.3.1 There are inherent characteristics of the rail environment that must be understood to recognize the need for many of the requirements identified in this publication.

1.3.2 Forces encountered within the rail vehicle are induced by shock and/or vibration. In most instances, the force is a complex result of both shock and vibration. Force input due to shock is mainly a result of impacts during switching and train slack action (run-in and run-out during train movement). Force input due to vibration is a result of the movement of the railcar's wheels on the rails. This vibration force can act either in a vertical or lateral plane. These forces are due to the movement of the car wheels on the rails, the truck geometry, rail joints, rail elasticity, nonuniformities of the rail and wheels, and overall track condition. When all these factors are acting on a rail vehicle, the resultant force is very complex.

1.3.3 The lading in a rail vehicle can also translate forces from one direction to another. Loading patterns that provide lengthwise contact between rolls will tend to transfer all longitudinal train action forces into the car endwall and the last roll at the endwall.

1.3.4 Loading patterns that do not provide lengthwise contact between rolls will transfer some longitudinal train action forces into the car sidewalls, including doors.

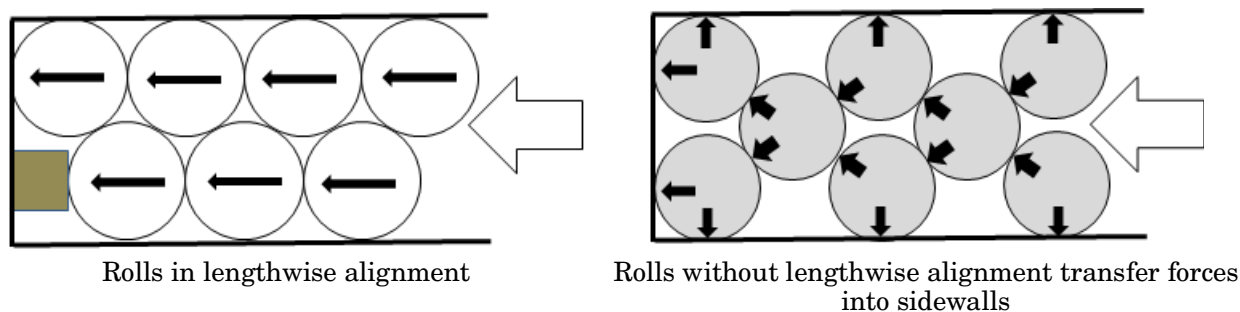


Figure 1.1 Load forces

1.3.5 Uncontrolled movement and/or displacement of the lading in a rail vehicle can cause safety problems, equipment failure, damage, and unloading problems. The following minimum loading standards in conjunction with proper packaging will provide safe arrivals.